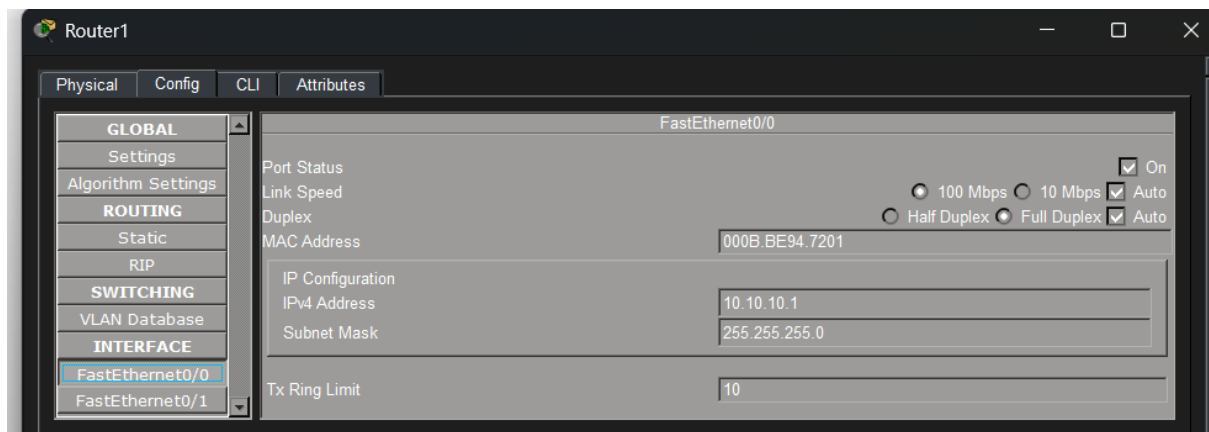


The screenshot shows a window titled "Router1" with a dark theme. The window has a sidebar on the left with a tree view containing the following categories: GLOBAL (expanded), Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, INTERFACE, FastEthernet0/0, and FastEthernet0/1. The main area is titled "Global Settings" and contains the following fields and buttons:

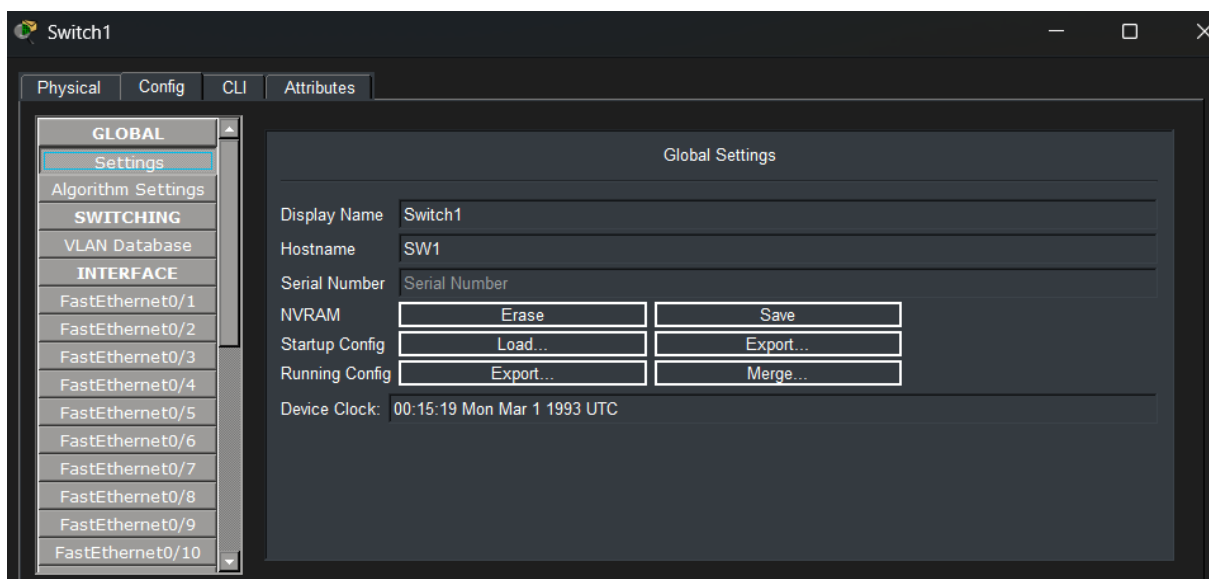
- Display Name: Router1
- Hostname: R1
- NVRAM: [Erase] [Save]
- Startup Config: [Load...] [Export...]
- Running Config: [Export...] [Merge...]
- Device Clock: 00:14:13 Mon Mar 1 1993 UTC

At the bottom of the window, there is a section titled "Equivalent IOS Commands" which is currently empty. Below this section, the text "Press RETURN to get started!" is displayed.

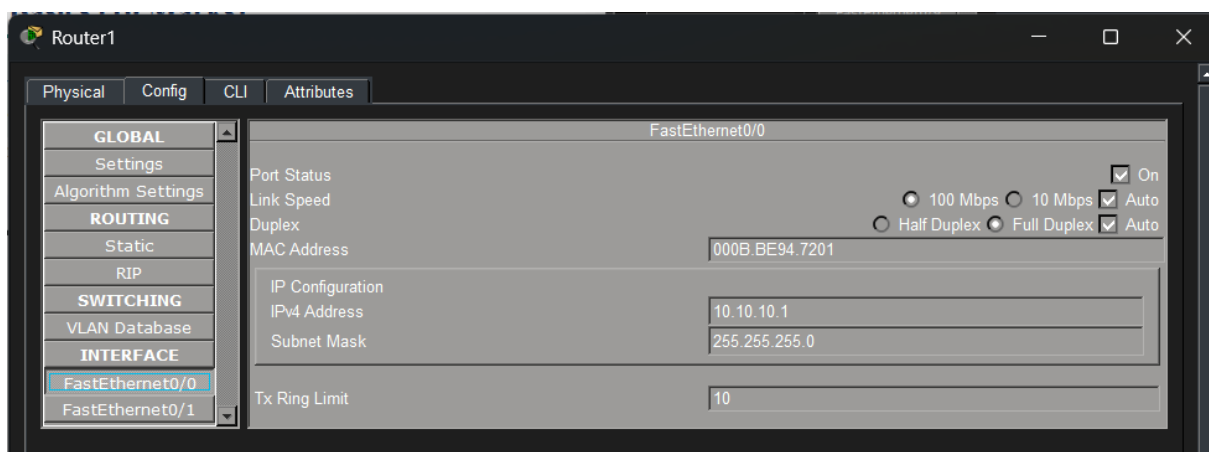
29.



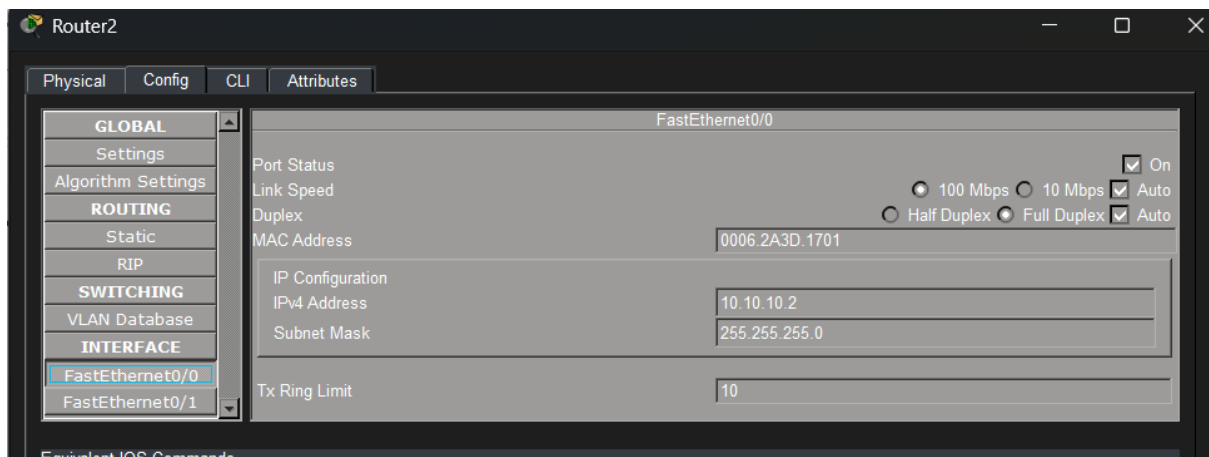
30.



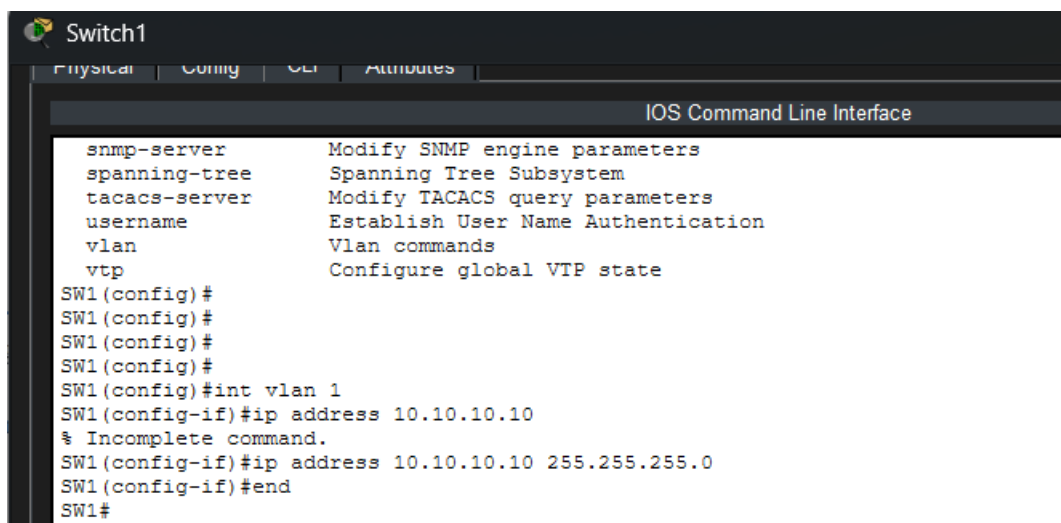
31.



32.



33.



34.

```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface

SW1#con t
% Ambiguous command: "con t"
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int vlan 1
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW1#
```

Top

36. Found R2, disabled R1's CDP as per instructions on 37.

```
SW1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme  Capability  Platform  Port ID
R2                Fas 0/2        130      R           C2800     Fas 0/0
SW1#
```

37. Disabled R1's CDP.

```
Router1

% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip int brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.10.10.1      YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#fastE0/0
      ^
% Invalid input detected at '^' marker.

R1(config)#Int fastEthernet F0/0
      ^
% Invalid input detected at '^' marker.

R1(config)#Int fastethernetF0/0
      ^
% Invalid input detected at '^' marker.

R1(config)#Int fastEthernet0/0
R1(config-if)#no cdp enable
R1(config-if)#
```

38.

```
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#
```

39.

```
SW1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce Holdtme  Capability Platform  Port ID
R2                Fas 0/2      130      R           C2800      Fas 0/0
SW1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce Holdtme  Capability Platform  Port ID
R2                Fas 0/2      170      R           C2800      Fas 0/0
SW1#
```

40.

```
R2#show ip int brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.10.10.2      YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R2#
```

41.

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#shutdown

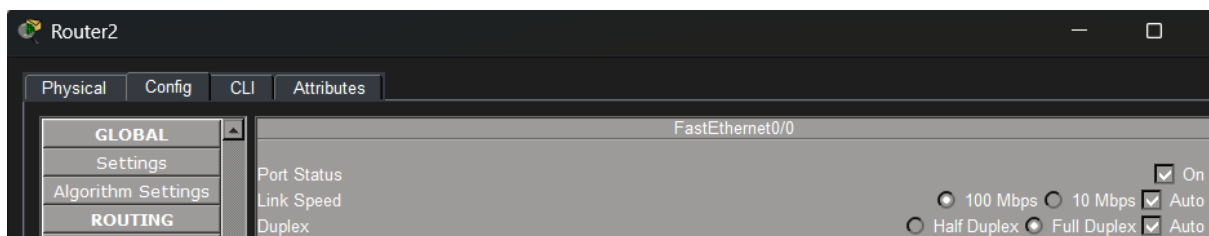
R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip int brief
Interface      IP-Address      OK? Method Status                Protocol
FastEthernet0/0 10.10.10.2      YES manual administratively down down
FastEthernet0/1 unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#
```

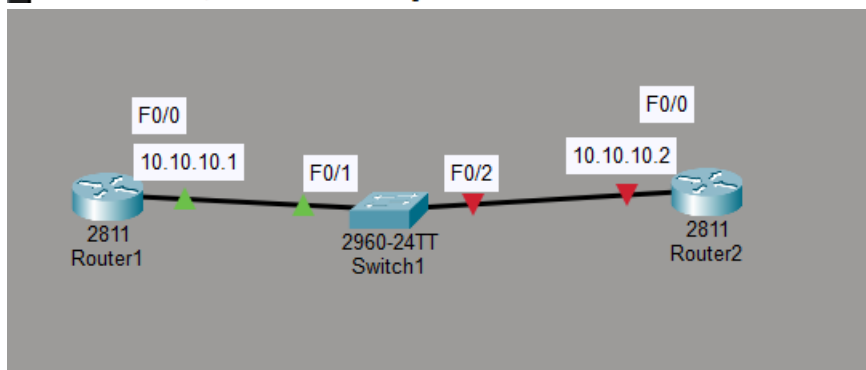
42.



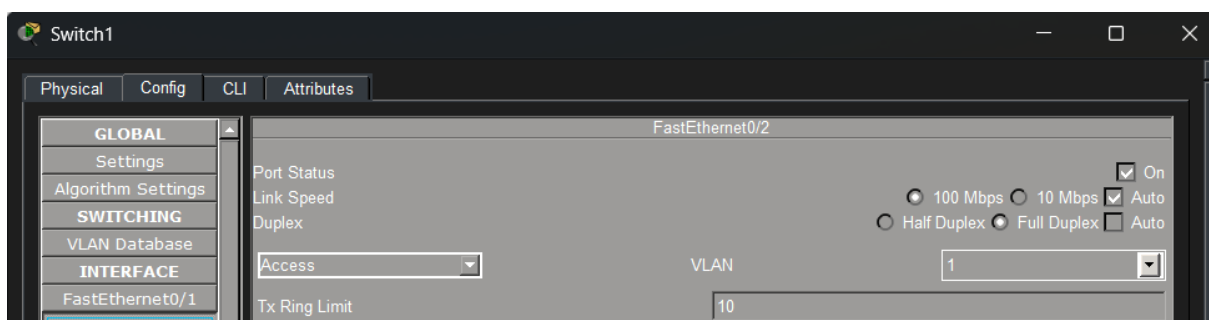
43. Interface went down with the duplex mismatch.

```
SW1#show ip int brief
Interface      IP-Address      OK? Method Status                Protocol
FastEthernet0/1 unassigned      YES manual up
FastEthernet0/2 unassigned      YES manual down

```

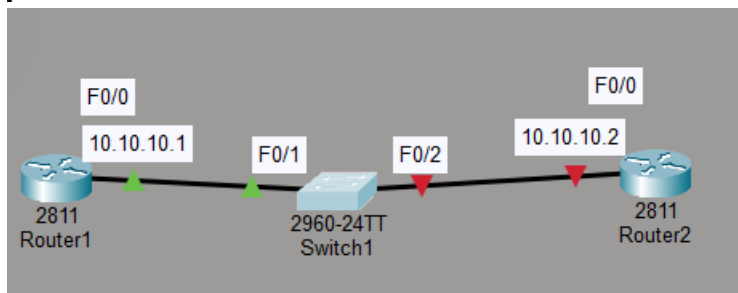


44.



45. Interface is down when the speed is different on both interfaces.

```
SW1#show ip int brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/1    unassigned      YES manual up          up
FastEthernet0/2    unassigned      YES manual down        down
```



46.

```
R2#show ip int brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    10.10.10.2      YES manual up          down
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2#
```

R2 int is still up, but it is down on the switches side.

Reflection question

47. The duplex mismatch causes the interfaces to be down, they must match to work. As soon as I changed them back it was okay and back to normal, so no lasting effects on the devices.

The duplex setting is basically changing if the device can send and receive data at the same time, if devices don't match, transfers would collide. So it shuts down.